

Bachelor Thesis

Blockchain Scalability

Optimizing Committee Selection for Sharding

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Abstract

This project is focused on exploring how to optimize blockchain sharding protocols, where the basic idea is to parallelize the ledger protocol. The work emerges from two protocols; a sharding protocol, GearBox, and a consistency protocol, Consistency-or-Die. Applying an analysis that accounts for inactive parties in sharding committees results in more optimistic committee sizes, and explores a rarely addressed aspect in existing sharding protocols.

In a network with 10,000 parties, 60% honest, 20% corrupt and 20% inactive, the new analysis compute a *Quorum* which is approximately one-third the size of the existing GearBox committee, $\frac{241}{717}$ having $Safety = \frac{1}{3}$.

This project demonstrates a potential improvement to GearBox's committee analysis, yet before implementing it in practice, further examination of committee overlap is required.

I would like to express my sincere gratitude to my supervisor, Bernardo. Without his guidance this project would not have been possible.

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Chapter 1

Introduction

Since the publication of the blockchain whitepaper, cryptocurrencies, and especially since the 2017 blockchain boom, blockchain has gained notable popularity and moved from being a niche cryptographic experiment into a foundational technology. Blockchain is a method for storing transactions securely. Transactions are collected into *blocks* and with a link to its previous block, it forms a *chain*, hence the name. In classic blockchain, every party processes every transaction, however this is unnecessary and exactly the challenge blockchain sharding intent to solve. As adoption of blockchain grows, so does the demand for being able to scale the systems. In blockchain protocols, consensus is used to determine which transactions are accepted and their order. Sharding introduces multiple smaller parallel committees from the group of participants, where each shard must run its own consensus to validate transactions. With this approach, shards share the workload while running in parallel. Shard efficiency is closely linked to the size of the committee size - that is, the number of parties in a shard. One sharding protocol, GearBox, tries to solve this by dynamically change shard size, but allows shards to lose liveness. In a real world setting, participants may go offline the network and therefore become inactive, others actively behave maliciously and of course many will be honest. Nevertheless, few sharding analyses consider inactive but not malicious parties.

This project focuses on analysing and experimenting with committee selection methods that pursue the production of more efficient shards while preserving safety of the shards. A more efficient sharding protocol ultimately results in more scalable blockchain implementations.

More formally, this project will dive into; *How to form smaller committees having inactive, malicious and honest parties while preserving safety and pursuing better liveness of the sharding protocol in realistic situations.*

Chapter 2

Background

2.1 Blockchain

Blockchain is a technique to store information securely, often being transactions, in a decentralized and verifiable way. The two most adapted ways to create blockchains are Proof-of-Stake (PoS) and Proof-of-Work (PoW). This project continues working with PoW protocol, where miners build blocks by solving a computational problem. PoS is another protocol, used in Ethereum among others, using coins as a safety to become a validator. In the PoW protocol, transactions are grouped into *blocks* where each block is linked to the previous block, forming an immutable "chain" hence the name *blockchain*. [6, page 27,]

Transactions are often structured using a Merkle tree [3, section 3.2], which produces a single *root hash* by recursively hashing all transactions in a specific binary pattern. This root hash, with the previous block's hash, timestamp, and a special value *nonce* is included in the block.

The nonce is the central point for the process of creating the Proof of Work. To create a valid block, validators (also known as miners) must find a nonce that, together with the rest of the block header, produces a hash that meets a difficulty target. The target typically requires the resulting hash to start with a certain number of zeros.[7, section 5] Since cryptographic hash functions are built to be unpredictable, the only practical way to find a valid nonce is by brute force.

2.2 Sharding

Sharding has traditionally been used to split large datasets into smaller parts called shards, making it simpler for more parties to work on large dataset and databases in parallel. [4, page 4]

In the context of blockchain, a similar approach is used for sharding, however instead of partitioning datasets, sharding in blockchain partitions consensus work by assigning parties into smaller groups called shards. Splitting the blockchain consensus into shards, increases the capacity of the blockchain by approximately the same multiple as the number of shards.[4, page 4]

Consensus is needed in the process of validating transactions across the network, and which blocks to add to the chain. [8, section 2.2]

The advantage of sharding is obvious when imagining a new transaction that needs to be verified. It does not make sense to run every transaction through every party to validate the transaction; it is much more efficient to run the transaction through all parties in a smaller shard. Thus, multiple shards can handle more transactions at once.

Another aspect of sharding protocols that are important to account for is re-sampling. If a shard does not progress, or if some upper time limit is exercised for a shard, the shard is stopped and another shard is created. All parties of this new shard needs to download and validate the whole ledger before starting to build upon the ledger. This takes time where the shard could potentially progress.

The internal efficiency of the shards depend also on their size, smaller shards are more efficient because they have less overhead such as message exchange during the consensus protocol and simply less parties to wait for. However, smaller shards are easier to corrupt, requiring larger shards, which introduces conflicting desires.

2.3 GearBox Protocol

GearBox is an effective addition to blockchain sharding, that leverages the sharding safety-liveness dichotomy, allowing shards to lose liveness while remaining safe.

Safety is defined as parties in the shard agree on the sequence of messages.

Liveness is defined as the shard will eventually output new messages to all parties.

Safety and liveness are bounded by thresholds in the paper relative to the *safety-liveness dichotomy*, which says that for partially synchronous protocols safety and liveness must satisfy the condition $2L + S < 100\%$. Partially synchronous is defined as all messages are delivered but with unknown delay, i.e. we do not have to worry about messages being dropped but there may be huge latencies.

This is the most realistic setting for real life.

GearBox proposes different ways to realize *how to select the committee*, when the committee size is established, however they all build upon the concept of having a source of randomness. The work of this project assumes working with the simple protocol presented in GearBox. [2, page 14] As the method of actually selecting committee members does not affect the size of the committees, this project will not further unfold this.

A shard is instantiated with fixed safety and liveness thresholds that remain constant for as long as it runs. As long as these thresholds are not broken, the shard is safe and live except with negligible probability. If a shard is to lose liveness, the shard is stopped and then respawned with new parameters having higher liveness threshold. This results in a dynamically adjusting in shard sizes to get as close to the actual needed shard size, which allows much smaller shards in practice.

Other sharding protocols have to guarantee both liveness and safety under the worst-case corruption. In that case is it unknown whether an adversary tries to break liveness or safety, why it must consider equal thresholds, which greatly impacts the committee size. [2, page 3]

2.4 Consistency or Die Protocol

Consistency-or-Die (CoD) is a protocol for protecting key transparency logs against so-called split view attacks. Very simplified, the protocol says; a user receives a view from the channel maintainer, asks the network whether the view corresponds with other users view. Other users need to endorse the view to approve its authenticity for the receiver to trust the view. CoD uses Verifiable Random Functions (VRF) that chooses who should endorse the view, which corresponds to selecting a committee in practice. To choose how many parties needs to endorse the view, CoD computes a *Quorum*.

2.4.1 Analysis

In CoD a quorum is computed; the size of a committee with given parameters. The quorum considers not only honest and corrupt parties, but furthermore *inactive* parties. This very property is vastly powerful, as we do not need to assume inactive parties are actively malicious, as GearBox conservatively does. There are many valid reasons for being inactive. We know for a fact that computers can be turned off, lose power, or simply not have a connection to the network. This is important because if a party is inactive due to one of many natural reasons, we do not need to consider this party as actively malicious, but rather just as a "sleepy" party that is unavailable.

2.4.2 VRF & Grinding

For the selection of committee members, CoD utilizes Verifiable Random Functions, VRF. In short, the approach consists of parties receiving a seed derived from the previous block. Parties run the VRF locally, and together with the seed and some unique secret key for that party, the VRF computes a pseudo-random value. A certain threshold establishes whether a party is selected in the committee, depending whether the pseudo random value from the VRF, is above or below that very threshold.

Grinding refers to the concept of adversarial parties run the VRF with many different inputs to learn the outputs of the VRF. CoD assumes, for worst case scenario, adversaries can tamper with the seeds used in the VRF to become tolerant to these kinds of attacks. The concern is that an adversary could tamper with the committee selection to ensure certain parties would be selected for the committee, due to the potential knowledge of the VRF's deterministic behaviour. This altering with the committee selection may ultimately corrupt a shard. Worst case of a successful grinding attack is that all corrupt parties together with half of the honest parties will try to verify a fake certificate. To account for this, the CoD analysis weights the tail probabilities using $M + \frac{H}{2}$, M being the amount of malicious parties and H the amount of honest parties.

Chapter 3

Committee Size Analysis

Based on the strengths and weaknesses of the GearBox and CoD protocol, it is evident that it would be interesting to try to implement the analysis from CoD into a setting for blockchain sharding committees like GearBox to consider inactive parties in the sharding setting.

3.1 Adjustments and Considerations

Before proceeding with my work, we must clarify the assumptions and the environments we are exploring, incorporating adjustments from the other protocols.

3.1.1 Total Participants in the Pool

Various sources are working with around 10,000 parties (validators) in total, both GearBox and in a PoS based Ethereum protocol [5, section V, A] use this amount of total parties. Likewise, I have been working with the same assumption of 10,000 parties.

3.1.2 Security Parameter

Choosing the security parameter, λ , is a crucial part of the analysis. CoD considers security parameters from 30-256 and GearBox uses 60.

In this project, 60 has been considered a good number, as it provides high security even when considering running the protocol 1 million times.

Given $\lambda = 60$, the probability of failure when sampling the committee is 2^{-60} , therefore the probability for failing when sampling one million committees is

$$2^{-60} \cdot 10^6 = 2^{-60+\log_2(10^6)} = 2^{-60+19.934} = 2^{-40.066}$$

Assuming that shards will re-sample every week, it is possible to determine how long time shards can run with this probability of failure - and for how many shards. Table 3.1 illustrates these numbers.

Even after 1923 years, the probability of ever having had one failure across 10 shards remains at $2^{-40.066}$.

Shards	Weeks	Years
10	100,000	1923
15	66,666	1282
20	50,000	961
25	40,000	769

Table 3.1: CoD: Lifetime for 1 million samples and various no. shards. Computed using floor function & 52 weeks/year

3.1.3 Grinding - Perfect Randomness

Since the GearBox model assumes access to a global randomness beacon, this experiment follows the same unbiased source of randomness. As a result, grinding attacks, presented in section 2.4.2, which can arise using VRFs are not considered. In practice, this means instead of using the bound $M + \frac{H}{2}$ as CoD originally does, this experiment only considers M for number of corrupt parties and H for honest parties. These are used for computing the tails of the binomial distribution to ensure that having too few honest parties and too many corrupt parties in the committee are both negligible.

3.2 Computing

In an experiment, committee sizes are computed. It extends the code provided for statistical analysis of committee sizes in the original CoD paper. [1, reference 40] As presented, the code has been modified to no longer consider more than the accurate expected number of corrupted parties.

The experiment is executed using a total participant pool of 10,000. Furthermore, for the experiment, the honesty percentage is fixed for 60% as this is the least amount of honesty CoD calculates its quorum sizes for. [1, page 13, figure 3] The amount of actively malicious and inactive parties varies with 5 percentage point.

3.2.1 Code

To form the grinding change, the function *cod_cutoffs_batch* has been given slight extra functionality and an extra argument to not remove the functionality for grinding, but rather making it an option when running the function.

Running the adjusted code with my input values results in a long text file with following format:

```

...
p = 0.275599999...(expected committee size = 2756)  k = 1356
L: 0.000000000... - R: 0.000000000...
  log(L) = -59.81567957...
  log(R) = -60.41004133...

p = 0.275700000...(expected committee size = 2757)  k = 1356
L: 0.000000000... - R: 0.000000000...
  log(L) = -60.033247021...
  log(R) = -60.224208961....
...

```

Finding the best committee size simply involves identifying the smallest k (quorum) with both log values below -60 , which is λ , the security bit value. The example above is the result of 60% honesty and 40% corruption.

With this method, it is possible to obtain the quorums and their expected committee sizes for different levels of corruption, depicted in table 3.2 and as a graph in figure 3.1

Corruption Percentage	Expected Committee	Quorum
10	348	96
15	484	156
20	666	241
25	927	369
30	1299	560
35	1865	862
40	2757	1356

Table 3.2: CoD: Committee sizes tolerating various corruption levels

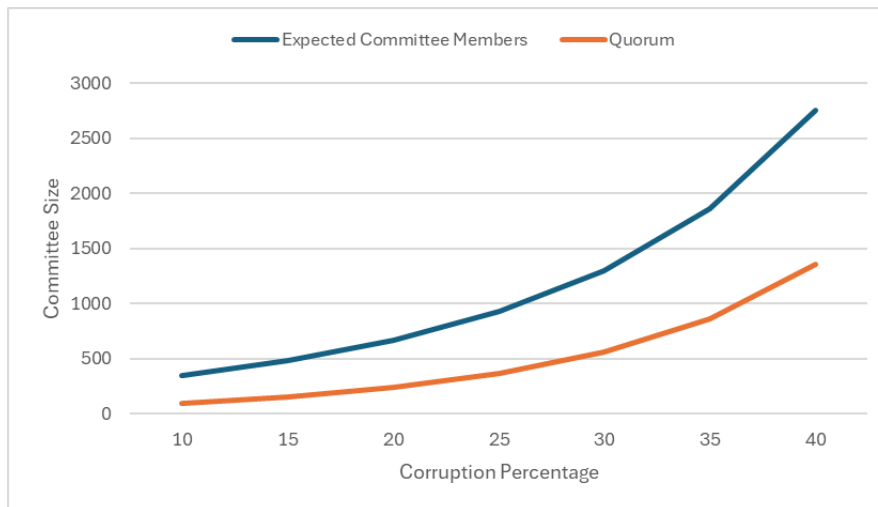


Figure 3.1: CoD: Committee sizes tolerating various corruption levels

In the GearBox paper, committee sizes for certain levels of corruption is computed, observe figure 3.2.

Liveness threshold:	0%	5%	10%	15%	20%	25%	30%	33.3333%
Safety threshold:	99%	89%	79%	69%	59%	49%	39%	33.3333%
n	t	Minimal committee size						
∞	30%	37	55	82	130	232	528	16 037
10 000	30%	35	51	75	116	207	462	5886
2000	30%	34	50	71	112	190	382	1716
∞	20%	28	38	52	75	115	199	854
10 000	20%	26	34	47	67	104	178	717
2000	20%	26	34	46	66	99	164	540
our work							state of the art	

Figure 3.2: Table from the original GearBox paper

However, with these values, it is hard to compare how well the CoD analysis performed, since the honesty levels in GearBox are 70 and 80%.

Therefore, another experiment with the statistical analysis for the GearBox protocol was conducted with more comparable input arguments. The results can be found in figure 3.3 and table 3.3. As can be observed, it was infeasible to produce results for 35% and 40% corruption, as these would require several days to complete.

The computation is slow due to the use of exact combinatorial functions inside a loop. The program runs iteratively from 1 to n , computing the failure probability for every committee size until one fulfils the security parameter requirement. This becomes more noticeable for higher corruption levels, which require bigger committees resulting in more iterative computations.

This might be optimizable, however as this project does not focus on this analysis it has not been investigated.

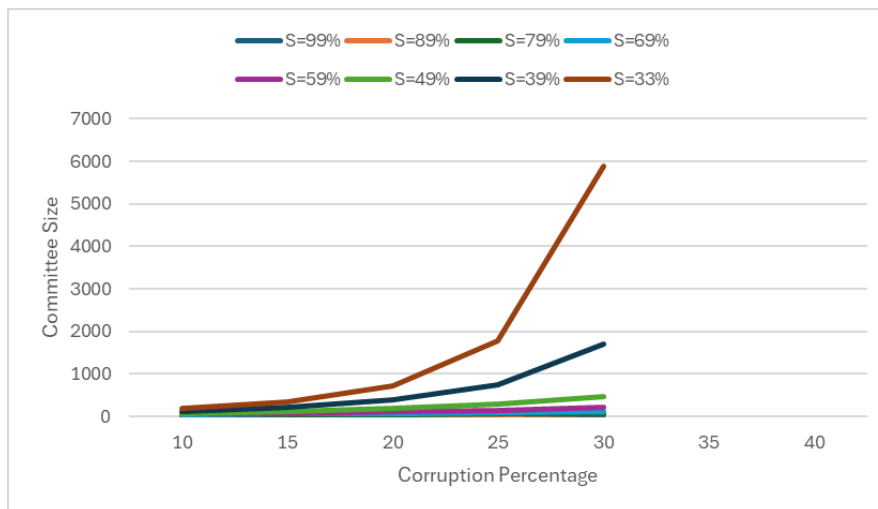


Figure 3.3: GearBox: Committee sizes tolerating various corruption levels and different safety guarantees

Safety - Liveness Corruption	S=99% L=0%	S=89% L=5%	S=79% L=10%	S=69% L=15%	S=59% L=20%	S=49% L=25%	S=39% L=30%	S=33% L=33%
10	19	23	28	38	51	76	126	183
15	22	27	37	51	73	119	218	348
20	26	34	47	67	104	178	385	717
25	30	42	60	87	146	278	744	1782
30	35	51	75	116	207	462	1713	5886

Table 3.3: GearBox: Committee sizes tolerating various corruption levels and different safety guarantees

Chapter 4

Discussion

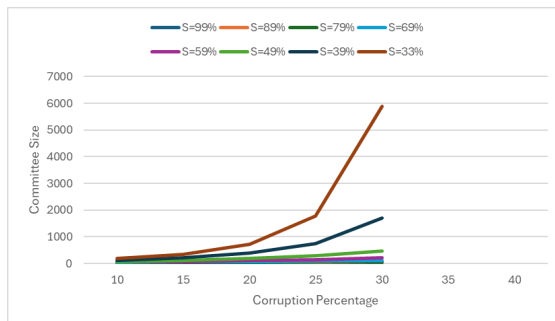


Figure 4.1: GearBox

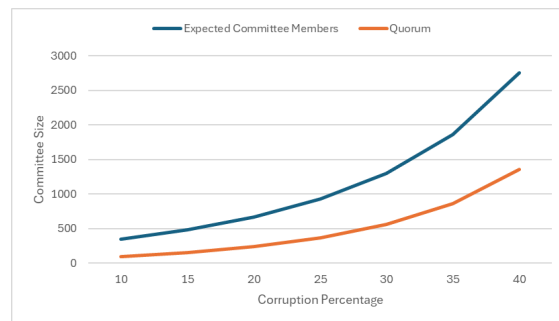


Figure 4.2: CoD

4.1 Results

Despite missing committee size results for the 35% and 40% mark for the GearBox analysis, it is quite clear that the CoD analysis generates more optimal committee sizes than the GearBox analysis that does not consider inactive parties. Committee sizes for the GearBox analysis also grow quickly, especially for lower safety thresholds, which nevertheless makes sense.

Unlike GearBox, the CoD analysis does not need to sacrifice liveness for safety, reducing the inevitable unnecessary re-sampling. This means less time spent re-validating the ledger and more time available for computation.

It is also worth noting that the safety analysis are hard to compare, as the CoD committee is both safe and live except with the small probability, whereas the GearBox analysis chooses to sacrifice liveness to keep the shard safe.

4.2 Corruption Tolerance

An interesting observation about the CoD's committee, is how much corruption and inactivity it tolerates. While it does make sense that more parties need to "speak" as corruption grows, it is interesting to observe how few parties are required to speak in the expected committee.

Figure 4.3 shows the fraction $\frac{Quorum}{ExpectedCommittee}$ for different corruption levels, representing the share of parties that must contribute to the protocol. As this fraction is rather low, it shows that it tolerates a high amount of inactive or corrupt parties.

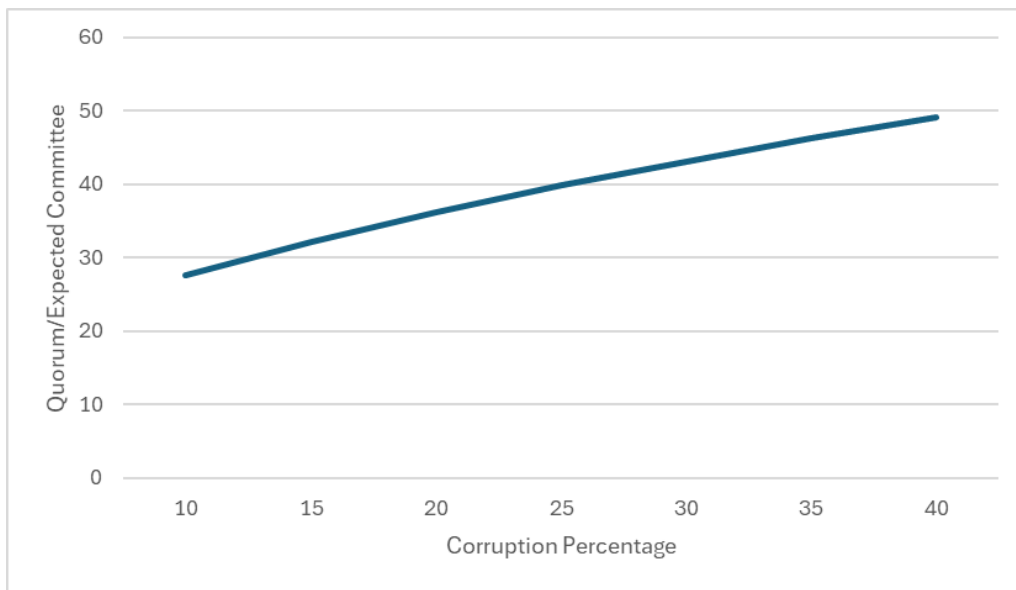


Figure 4.3: Quorum/Expected Committee fraction for corruption levels

4.3 Storage Optimizing

In the consensus protocol, information about which parties validated are saved in the blocks. In practice methods differ a bit, but the point is the information is stored. An obvious advantage is that when fewer parties are needed to reach consensus, the storage required naturally decrease. Having even smaller committee sizes acts as yet a positive side effect, as when we need fewer parties to validate a block there is less information to store. The result of this is more efficient storage of the blockchain and potentially optimization of the initial ledger validation process.

4.4 Committee Overlap

Another factor that would be interesting to explore, is how well the total party pool is covered when assigning parties to shards. With this committee selection technique, some overlap will happen, meaning one party will be part of more than just one shard. A big overlap is not a desirable feature as this undermines the purpose of sharding. If a party works in multiple shards that party becomes slower and therefore slows down both shards. This could easily become a limiting factor for running multiple shards.

4.5 Alternative Applications

As these experiments regards committees, and are not restricted to the domain of blockchain sharding, the theory is applicable in any circumstance using committees. One application could be Multi Party Computation using You-Only-Speak-Once model. This version of a MPC uses random committees which could be chosen in a similar approach as the committees in this project.

This project did not investigate other domains further, nevertheless it remains interesting to optimize other fields.

4.6 Considerations

Bringing the different aspects together, we see from the results that the new CoD analysis does result in smaller Quorums - and in some cases smaller expected committee sizes. Ones choice of analysis should reflect how conservative the protocol should be - if you have some data or are confident in corruption assumptions it is definitely possible to use the new CoD analysis instead of GearBox's more conservative analysis.

This choice in some sense affects the amount of overhead communication, bandwidth, and decision time - which might be worth considering before drawing a conclusion.

Chapter 5

Conclusion

Optimized blockchain sharding leads to more scalable solutions as it allows the network to process more transactions, without having all parties handle the whole workload.

The experiment generated committee sizes that were significantly smaller than the ones discovered before. For 20% corruption the experiment generated Quorum roughly one-third the size of that generated by GearBox's analysis, $\frac{241}{717}$ having $S = \frac{1}{3}$. As presented and discussed this affects both the consensus efficiency of the shard and the storage requirements, and may even optimize future work related to tracking or validating block participants.

As discussed, it would be interesting to keep working with how parties will overlap in committees as this project did not reach that far, however this is an important aspect to look into to determine if this work would be useful in sharding protocols. If too much overlap arises, parallelism is limited by the few parties who has to act in more than one shard, causing potential bottlenecks.

If future experiments confirm low overlap, this analysis has the potential to be implemented in sharding protocols, and should contribute to more efficient execution of such sharding protocols - ultimately resulting in more optimal blockchain execution.

The question regarding conservative protocols should be discussed as well; optimized protocols are often welcomed as they result in lower operation costs but in aspects like these, security is a central priority.

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